# Exact Least Concatenation Lengths for Two Infinite Families in OEIS A069862

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**Abstract**

For OEIS A069862, let $a(n)$ be the least positive $k$ for which the decimal concatenation of $n+1, \ldots, n+k$ is divisible by $n$. We prove the exact formula $a(n) = n$ whenever $n \mid 10^{\ell(n+1)}$, and the distinct unbounded family $a((10^{6j+3}+1)/7) = 2(10^{6j+3}+1)/7 - 1$ for all $j \geq 0$. We also establish a constant-digit-width reduction for the cases $10^D \equiv 0, 1, -1 \pmod{n}$, giving a further prime-power criterion. Each identity proves both existence and *minimality* on its specified family; none proves existence for every positive $n$. The known classification of the indices satisfying $a(n) = 2$ was already recorded by David Wasserman (2005) in OEIS A069860 and is explicitly excluded from our novelty claims.

## 1 Definition and a modular recurrence

Let $\ell(t)$ denote the number of base-10 digits of $t \geq 1$. Define

$$C_n(0) = 0, \qquad C_n(k) = 10^{\ell(n+k)} C_n(k-1) + (n+k) \quad (k \geq 1).$$

Whenever $C_n(k) \equiv 0 \pmod{n}$, the sequence value $a(n)$ is the least such $k$. For $n > 1$, $C_n(1) \equiv 1 \pmod{n}$, and $a(1) = 1$.

**Proposition 1** (Constant-width reduction)**.** *Put $D = \ell(n+1)$ and $L = 10^D - n - 1$. For $1 \leq k \leq L$, write $R_k = C_n(k) \bmod n$. Then all the concatenated blocks have $D$ digits. Consequently,*

$$\begin{array}{lcll} R_k & \equiv & k & (10^D \equiv 0 \pmod{n}), \\ R_k & \equiv & k(k+1)/2 & (10^D \equiv 1 \pmod{n}), \\ R_k & \equiv & \lceil k/2 \rceil & (10^D \equiv -1 \pmod{n}). \end{array}$$

*The triangular expression is an integer before reduction, even for even $n$.*

*Proof.* In this range $n + i < 10^D$ and $n + i \geq n + 1 \geq 10^{D-1}$, so the recurrence modulo $n$ is $R_k \equiv BR_{k-1} + k$ with $B = 10^D$ and $R_0 = 0$. When $B \equiv 0$, the result is $k$; when $B \equiv 1$, it is $\sum_{i=1}^{k} i = k(k+1)/2$. When $B \equiv -1$, induction gives $R_{2t} \equiv t$ and $R_{2t+1} \equiv t + 1$. □

## 2 First exact family: a last-block principle

**Theorem 1** (Decimal-smooth moduli)**.** *For any $n \geq 1$ with $D = \ell(n+1)$, if $n \mid 10^D$, then*

$$\boxed{a(n) = n}.$$

*In particular $a(10^r) = 10^r$ for all integers $r \geq 0$.*

*Proof.* Every $n + k$ for $k \geq 1$ has at least $D$ digits, so $n \mid 10^{\ell(n+k)}$. The recurrence thus yields $C_n(k) \equiv k \pmod{n}$ *for all* $k \geq 1$, including after any digit-width transitions. All $1 \leq k < n$ are nonzero modulo $n$, while $k = n$ is zero. □

**Corollary 1.** *The formula holds whenever* $n = 2^u 5^v$ *and* $\max(u, v) \le \ell(n+1)$. *Examples are* $a(20) = 20$, $a(25) = 25$, $a(125) = 125$, *and* $a(1000) = 1000$.

## 3 Second exact family: an alternating residue

**Theorem 2** (An infinite family with $a(n) = 2n-1$)**.** *For every integer* $j \ge 0$ *let* $D = 6j+3$ *and* $n = (10^D+1)/7$. *Then* $n$ *is a positive integer and*

$$\boxed{a\left(\frac{10^{6j+3}+1}{7}\right) = \frac{2(10^{6j+3}+1)}{7} - 1}.$$

*In particular,* $a(143) = 285$.

*Proof.* Since $10 \equiv 3 \pmod 7$, $3^3 \equiv -1 \pmod 7$, and 3 has order 6 modulo 7, we have $7 \mid 10^D + 1$. For $D \ge 3$ and $n = (10^D+1)/7$, the inequalities

$$10^{D-1} \le n+1 \quad \text{and} \quad 3n - 1 < 10^D$$

hold. Hence all terms $n+1, \dots, n+(2n-1)$ have exactly $D$ digits. Moreover $10^D = 7n - 1 \equiv -1 \pmod n$. Proposition 1 now yields $C_n(k) \equiv \lceil k/2 \rceil \pmod n$ for every $1 \le k \le 2n-1$. The first value in this range whose ceiling is divisible by $n$ is precisely $k = 2n-1$. Thus no shorter prefix is divisible by $n$, and that prefix is. □

## 4 A companion exact criterion

**Proposition 2** (Odd prime powers)**.** *Suppose* $n > 1$ *is an odd prime power,* $D = \ell(n+1)$, $n \mid 10^D - 1$, *and* $10^D \ge 2n$. *Then*

$$\boxed{a(n) = n - 1}.$$

*For example* $a(9) = 8$ *and* $a(11) = 10$.

*Proof.* The hypotheses ensure $L = 10^D - n - 1 \ge n - 1$. The constant-width recurrence gives $C_n(k) \equiv k(k+1)/2 \pmod n$ for $k \le n-1$. Since $n = p^e$ with $p$ odd, $n$ divides $k(k+1)/2$ if and only if $p^e$ divides $k$ or $k+1$, because these consecutive factors are coprime. The first such positive $k$ is $n-1$. □

## 5 Prior art, verification, and limitations

The classification of *all* indices with $a(n) = 2$ follows from the two-block congruence $C_n(2) \equiv 10^{\ell(n+2)} + 2 \pmod n$. The equivalent complete formula was published as a formula for OEIS A069860 by David Wasserman in 2005. We do not present it as a new theorem or claim credit for that classification.

The dependency-free script `verify.py` checks the congruence identities, representative exact minima, and the published A069862 values where present. Finite checks only supplement the proofs. These results settle existence and the *least* admissible length on the stated infinite families, but do not establish all-index existence or an all-index closed formula for A069862. Broader literature priority remains unverified.

## References


[1] OEIS Foundation Inc., *A069862: Smallest k such that n divides the concatenation of numbers from (n+1) to (n+k)*, https://oeis.org/A069862 (accessed October 22, 2026).

[2] D. Wasserman, formula for *A069860: Numbers m that divide the concatenation of m+1 and m+2*, OEIS entry, August 25, 2005, https://oeis.org/A069860.